Logic Gates

The aim of this chapter is introducing all logic gates and their applications.The logic gates are the basic circuits of digital electronics systems just like resistors , capacitors in analog circuits.

The logic gate is an electronic circuit ,which has one or many inputs but only one output. The output is expressed in terms of binary code 0 or 1. It acts as switches in circuit that performs logical operations.Output depends on the logic gate and the input.Input is one of two states – high or low.Output is one of two states – high or low.Logic gates can be consturcted using simple switches,relays,vaccum tubes,transistors or diodes. Now due to their availability in the form of integrated circuits(IC's)they are used to construct digital circuits.

Before studying different types of logic gateslet us study few related terms

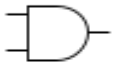
Logic: It is the statement, which gives condition for high output.Logic may be in terms of High/Low,True/False,1/0 etc.

Truth Table: It is table in which logic is expressed by taking all possible combinations of inputs and respective outputs.The number of possible combinations in atruth table is given by 2^n where n is no. of inputs.

Boolean Equation:It is mathematical equation representing particular gate for input and output.

Symbol: Each gate is represented by logical symbol .

Basic gates are AND gate , OR gate , Not gate



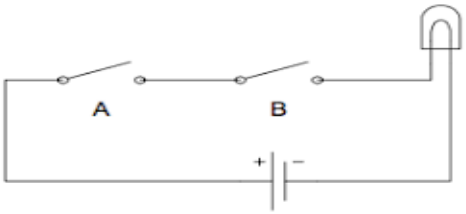
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(i) $A.0 = 0$ (ii) $A.1 = A$

(iii) $A.A = A$ (iv) $A.\overline{A} = 0$

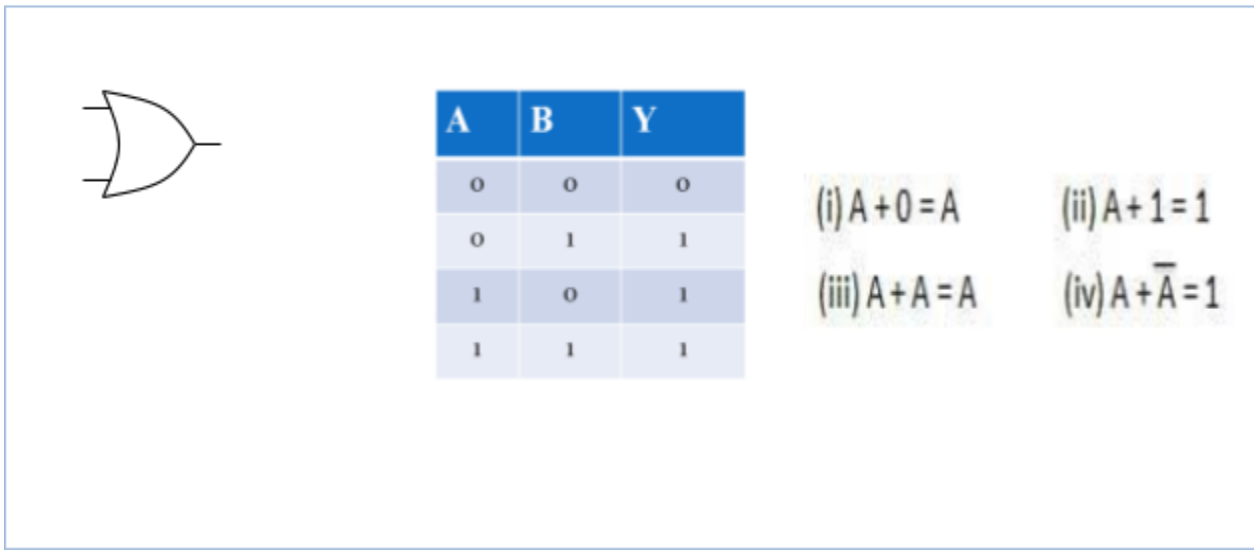
Logic :If A is true and B is true then Y is true. It means that the output of AND gate is high only when both inputs are high.

Equivalent Circuit:



A	B	LED
open	open	OFF
open	Closed	OFF
Closed	Open	OFF

Closed	**closed**	**ON**



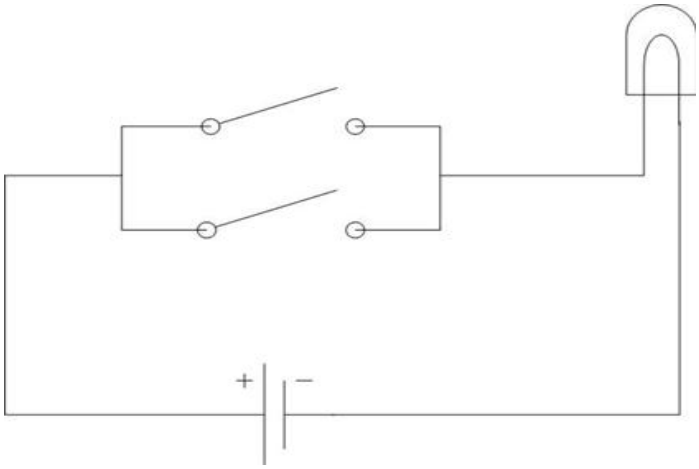
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(i) $A+0=A$ (ii) $A+1=1$

(iii) $A+A=A$ (iv) $A+\overline{A}=1$

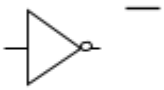
Logic : If A is True or B is True then Y is True

Equivalent Circuit:



A	B	LED
open	**open**	**OFF**
open	**Closed**	**ON**

Closed	**Open**	**ON**
Closed	**closed**	**ON**

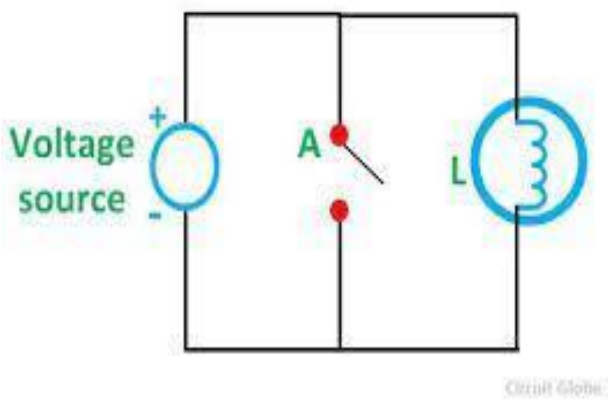


A	Y
0	1
1	0

$$\overline{\overline{A}} = A$$

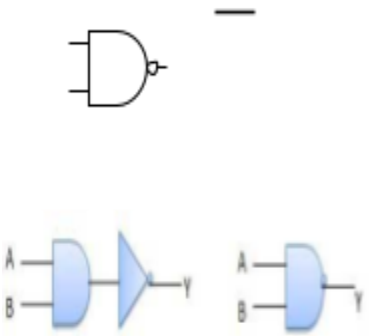
Logic : If A is true then Y is False and if A is false then Y is True

Equivalent circuit:



A	LED
open	**ON**

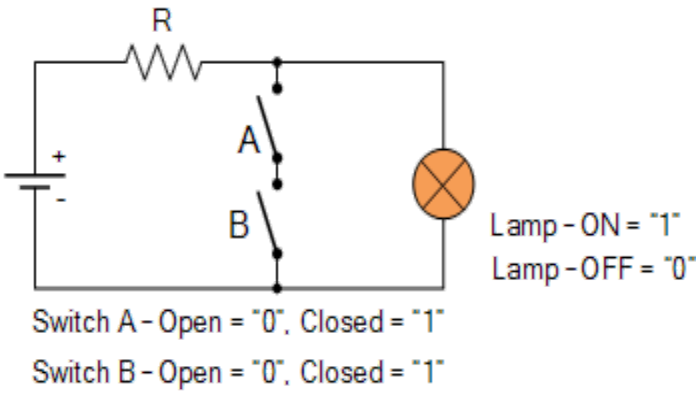
close	OFF



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

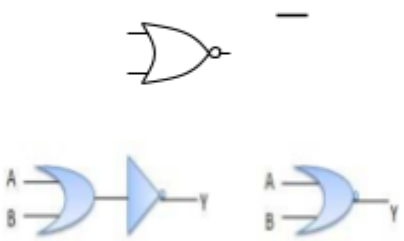
Logic: If A is high and B is high then Y is False

Equivalent Circuit:



A	B	LED
open	open	ON
open	Closed	ON

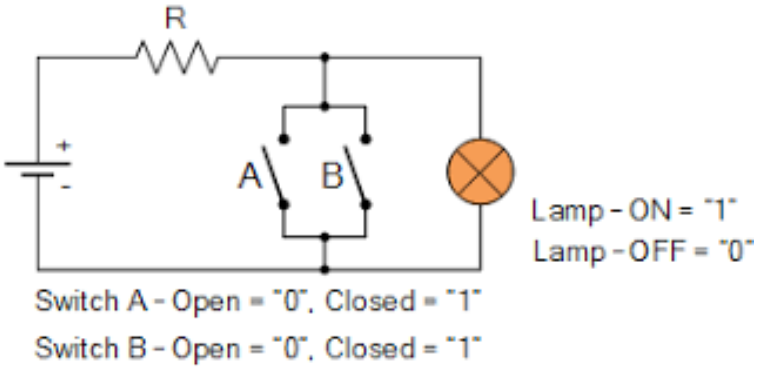
Closed	**Open**	**ON**
Closed	**closed**	**OFF**



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Logic : If A is True and B is True Y is false

Equivalent Circuit:



A	B	LED
open	**open**	**ON**
open	**Closed**	**OFF**
Closed	**Open**	**OFF**
Closed	**closed**	**OFF**



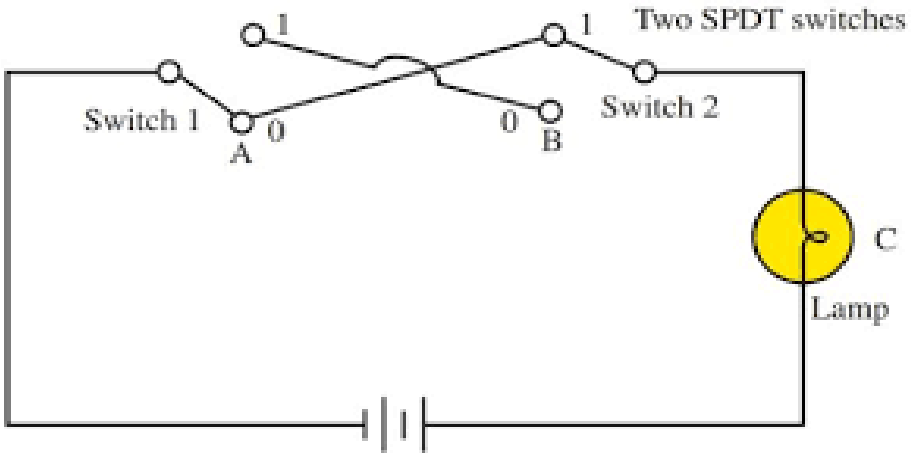
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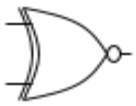
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Logic: When A is true or B is true but not both then Y is true

$Y=A\overline{B}+A\overline{B}$

Equivalent Circuit:



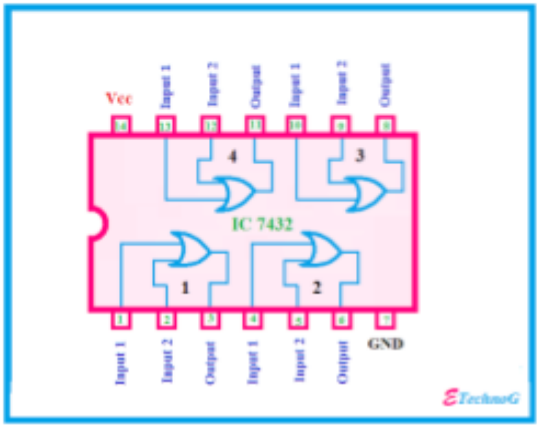


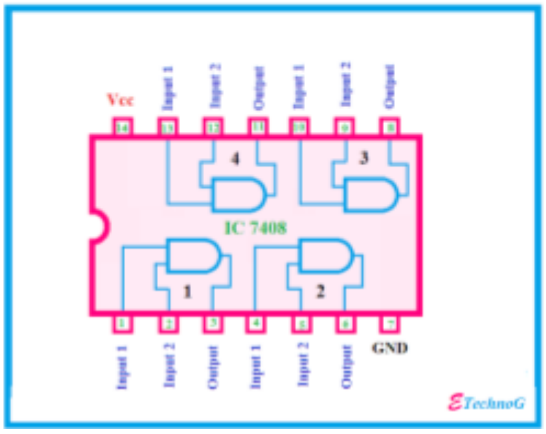


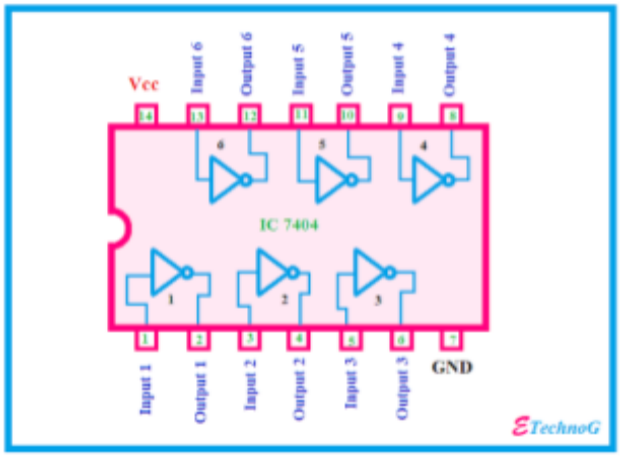
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

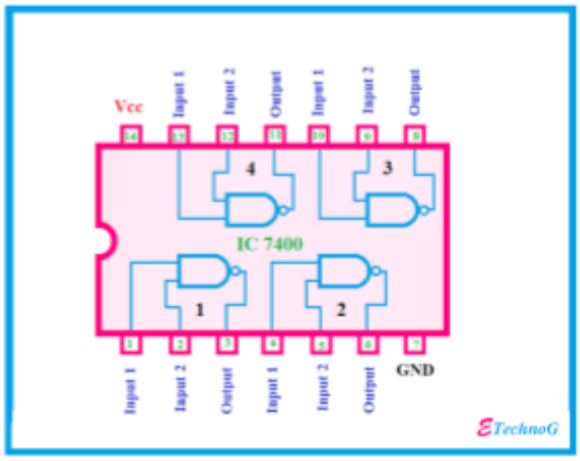
Logic: If A is true or B is true Y is False

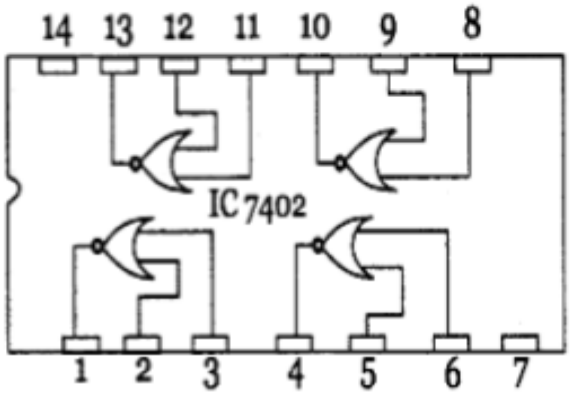
Ic 's used for different gates:

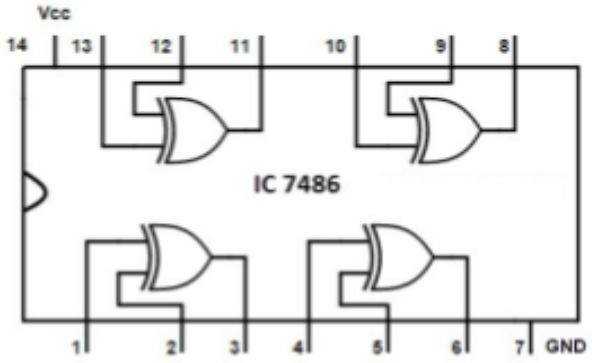


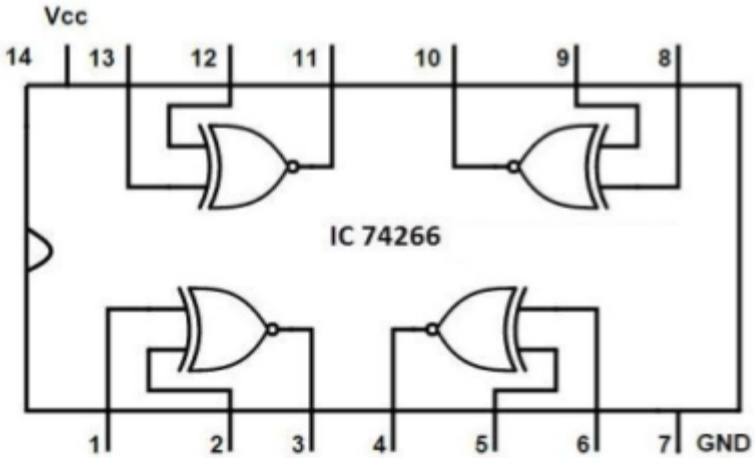



Vcc
Input 6
Output 6
Input 5
Output 5
Input 4
Output 4
IC 7404
GND
Input 1
Output 1
Input 2
Output 2
Input 3
Output 3
ETechnoG


Vcc
Input 1
Input 2
Output
Input 1
Input 2
Output
IC 7400
GND
Input 1
Input 2
Output
Input 1
Input 2
Output
ETechnoG


14 13 12 11 10 9 8
IC 7402
1 2 3 4 5 6 7


Vcc
14 13 12 11 10 9 8
IC 7486
1 2 3 4 5 6 7 GND





DIP-14



Applications of logic gates:

- In Microprocessor and Microcontroller circuits
- Used in push button switches
- Electronics and Electrical project circuits
- Embedded system applications

TTL and CMOS logic levels

